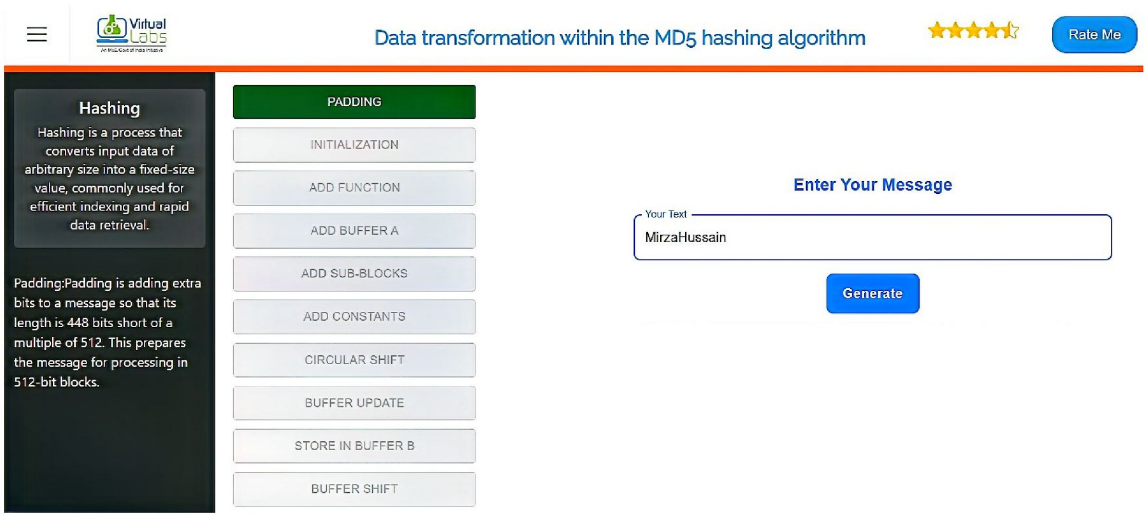
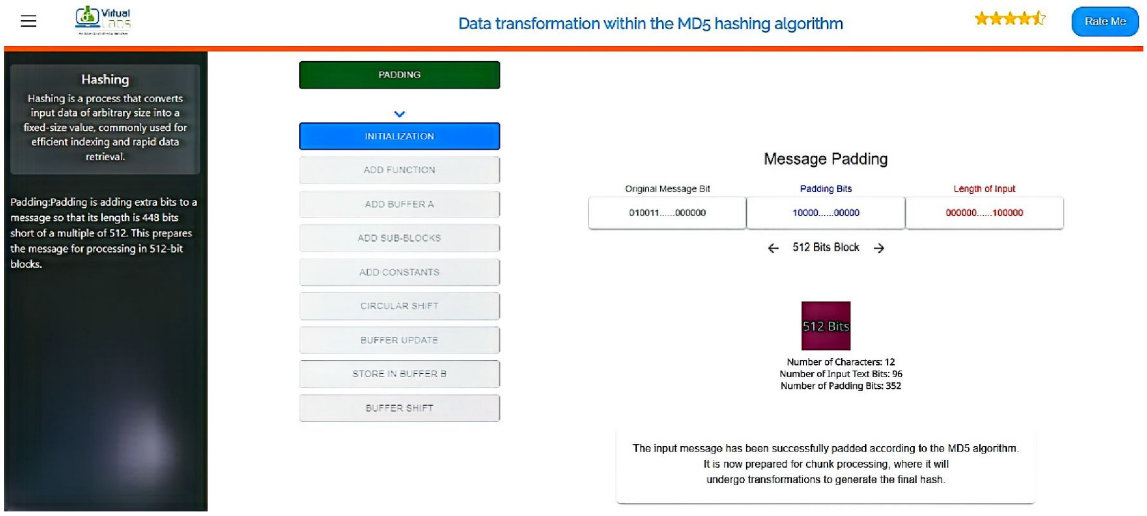
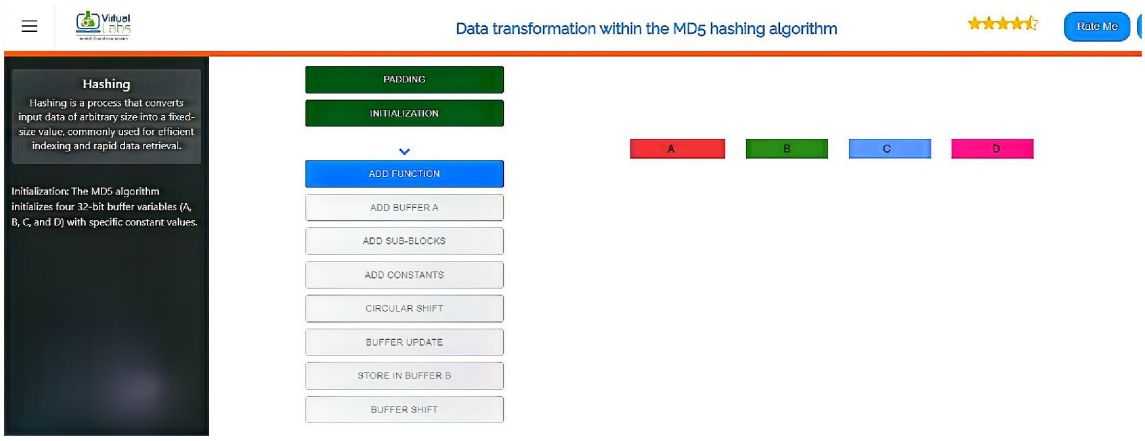
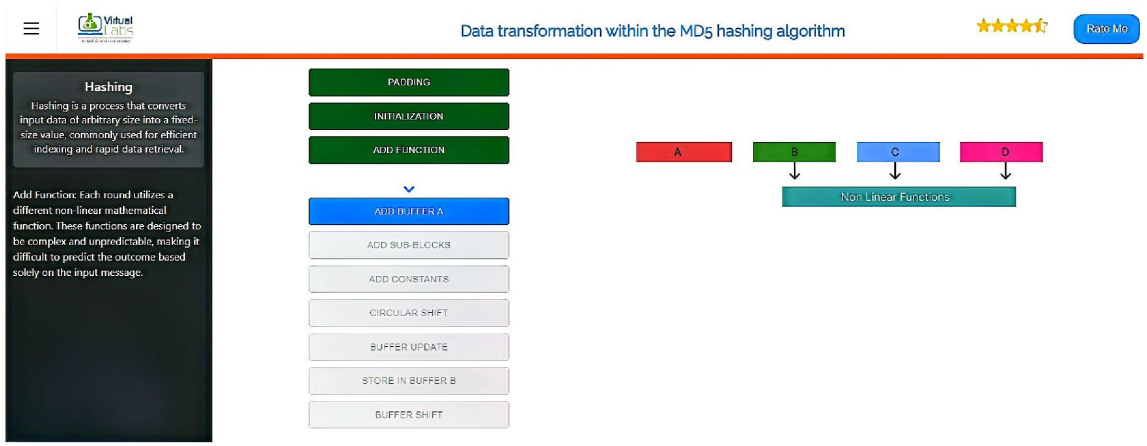

Data transformation within the MD5 hashing algorithm
Rate Me
Hashing
Hashing is a process that converts input data of arbitrary size into a fixed-size value, commonly used for efficient indexing and rapid data retrieval.
Padding:Padding is adding extra bits to a message so that its length is 448 bits short of a multiple of 512. This prepares the message for processing in 512-bit blocks.
PADDING
INITIALIZATION
ADD FUNCTION
ADD BUFFER A
ADD SUB-BLOCKS
ADD CONSTANTS
CIRCULAR SHIFT
BUFFER UPDATE
STORE IN BUFFER B
BUFFER SHIFT
Enter Your Message
Your Text
MirzaHussain
Generate


Data transformation within the MD5 hashing algorithm
Rate Me
Hashing
Hashing is a process that converts input data of arbitrary size into a fixed-size value, commonly used for efficient indexing and rapid data retrieval.
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PADDING
INITIALIZATION
ADD FUNCTION
ADD BUFFER A
ADD SUB-BLOCKS
ADD CONSTANTS
CIRCULAR SHIFT
BUFFER UPDATE
STORE IN BUFFER B
BUFFER SHIFT
Message Padding
Original Message Bit
Padding Bits
Length of Input
010011.....000000
10000.....00000
000000.....100000
512 Bits Block
512 Bits
Number of Characters: 12
Number of Input Text Bits: 96
Number of Padding Bits: 352
The input message has been successfully padded according to the MD5 algorithm. It is now prepared for chunk processing, where it will undergo transformations to generate the final hash.


Data transformation within the MD5 hashing algorithm
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Hashing
Hashing is a process that converts input data of arbitrary size into a fixed-size value, commonly used for efficient indexing and rapid data retrieval.
Initialization: The MD5 algorithm initializes four 32-bit buffer variables (A, B, C, and D) with specific constant values.
PADDING
INITIALIZATION
ADD FUNCTION
ADD BUFFER A
ADD SUB-BLOCKS
ADD CONSTANTS
CIRCULAR SHIFT
BUFFER UPDATE
STORE IN BUFFER B
BUFFER SHIFT
A
B
C
D


Data transformation within the MD5 hashing algorithm
Rate Me
Hashing
Hashing is a process that converts input data of arbitrary size into a fixed-size value, commonly used for efficient indexing and rapid data retrieval.
Add Function: Each round utilizes a different non-linear mathematical function. These functions are designed to be complex and unpredictable, making it difficult to predict the outcome based solely on the input message.
PADDING
INITIALIZATION
ADD FUNCTION
ADD BUFFER A
ADD SUB-BLOCKS
ADD CONSTANTS
CIRCULAR SHIFT
BUFFER UPDATE
STORE IN BUFFER B
BUFFER SHIFT
A
B
C
D
Non Linear Functions

Data transformation within the MD5 hashing algorithm

Rate Me

Hashing

Hashing is a process that converts input data of arbitrary size into a fixed-size value, commonly used for efficient indexing and rapid data retrieval.

Add Buffer A: Buffer A, which stores an intermediate hash state, is added to the current state to accumulate changes over multiple rounds, enhancing security.

- PADDING
- INITIALIZATION
- ADD FUNCTION
- ADD BUFFER A
- ADD SUB-BLOCKS
- ADD CONSTANTS
- CIRCULAR SHIFT
- BUFFER UPDATE
- STORE IN BUFFER B
- BUFFER SHIFT



Data transformation within the MD5 hashing algorithm

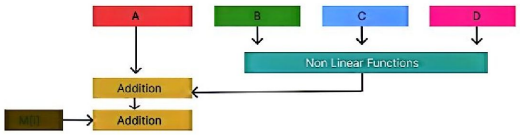
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Hashing

Hashing is a process that converts input data of arbitrary size into a fixed-size value, commonly used for efficient indexing and rapid data retrieval.

Add Sub-Blocks: The padded message is divided into 512-bit blocks, which are further split into sixteen 32-bit sub-blocks. These sub-blocks are then added to the buffer for processing

- PADDING
- INITIALIZATION
- ADD FUNCTION
- ADD BUFFER A
- ADD SUB-BLOCKS
- ADD CONSTANTS
- CIRCULAR SHIFT
- BUFFER UPDATE
- STORE IN BUFFER B
- BUFFER SHIFT



Data transformation within the MD5 hashing algorithm

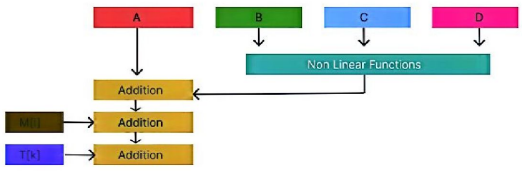
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Hashing

Hashing is a process that converts input data of arbitrary size into a fixed-size value, commonly used for efficient indexing and rapid data retrieval.

Add Constants: Predefined constants derived from the sine function are added to introduce randomness and strengthen security.

- PADDING
- INITIALIZATION
- ADD FUNCTION
- ADD BUFFER A
- ADD SUB-BLOCKS
- ADD CONSTANTS
- CIRCULAR SHIFT
- BUFFER UPDATE
- STORE IN BUFFER B
- BUFFER SHIFT



Data transformation within the MD5 hashing algorithm

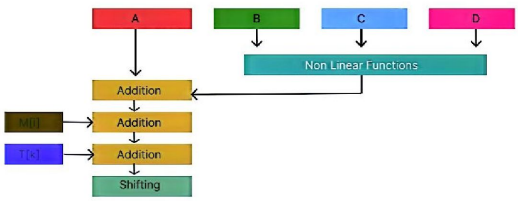
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Hashing

Hashing is a process that converts input data of arbitrary size into a fixed-size value, commonly used for efficient indexing and rapid data retrieval.

Circular Shift: A circular left shift operation is performed on buffer values to spread changes and improve diffusion

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- INITIALIZATION
- ADD FUNCTION
- ADD BUFFER A
- ADD SUB-BLOCKS
- ADD CONSTANTS
- CIRCULAR SHIFT
- BUFFER UPDATE
- STORE IN BUFFER B
- BUFFER SHIFT



Data transformation within the MD5 hashing algorithm

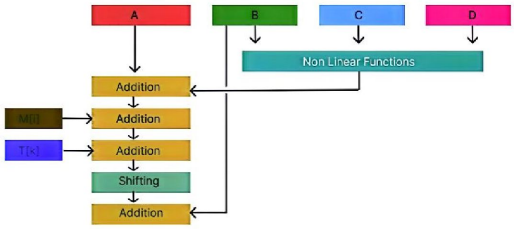
Rate Me

Hashing

Hashing is a process that converts input data of arbitrary size into a fixed-size value, commonly used for efficient indexing and rapid data retrieval.

Buffer Update: The buffer values (A, B, C, and D) are updated based on the processed sub-block, modifying the hash state.

PADDING
INITIALIZATION
ADD FUNCTION
ADD BUFFER A
ADD SUB-BLOCKS
ADD CONSTANTS
CIRCULAR SHIFT
BUFFER UPDATE
STORE IN BUFFER B
BUFFER SHIFT



Data transformation within the MD5 hashing algorithm

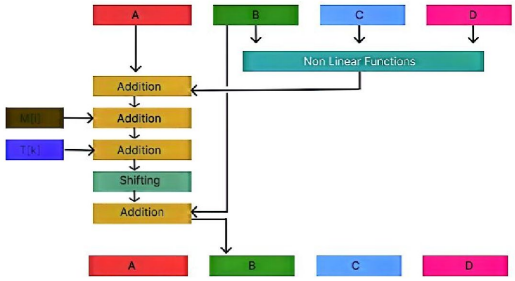
Rate Me

Hashing

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Store in Buffer B: The updated result is stored in Buffer B for the next step

PADDING
INITIALIZATION
ADD FUNCTION
ADD BUFFER A
ADD SUB-BLOCKS
ADD CONSTANTS
CIRCULAR SHIFT
BUFFER UPDATE
STORE IN BUFFER B
BUFFER SHIFT



Data transformation within the MD5 hashing algorithm

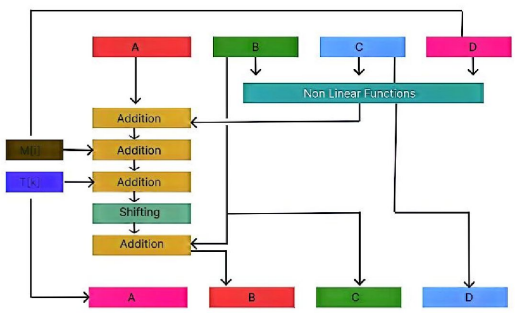
Rate Me

Hashing

Hashing is a process that converts input data of arbitrary size into a fixed-size value, commonly used for efficient indexing and rapid data retrieval.

Buffer Shift: Shift the buffer for the next iteration. This completes one cycle. This process will continue 16 times for each non linear function.

PADDING
INITIALIZATION
ADD FUNCTION
ADD BUFFER A
ADD SUB-BLOCKS
ADD CONSTANTS
CIRCULAR SHIFT
BUFFER UPDATE
STORE IN BUFFER B
BUFFER SHIFT



SHOW MD5 HASH VALUE

Data transformation within the MD5 hashing algorithm

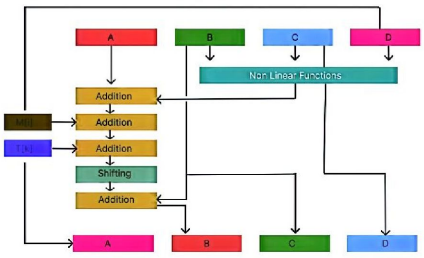
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ADD FUNCTION
ADD BUFFER A
ADD SUB-BLOCKS
ADD CONSTANTS
CIRCULAR SHIFT
BUFFER UPDATE
STORE IN BUFFER B
BUFFER SHIFT



SHOW MD5 HASH VALUE

836c8e3996938898b45e57ff97c982cfe

The MD5 hashing process is complete. Your hashed value is displayed above.